Mechanické (externí) paměti

19. otázka

19.

Mechanické pamětí

- Harddisk
- CD-ROM
- CD-RW
- DVD

Praktická ukázka

- Sestavte bistabilní klopný obvod

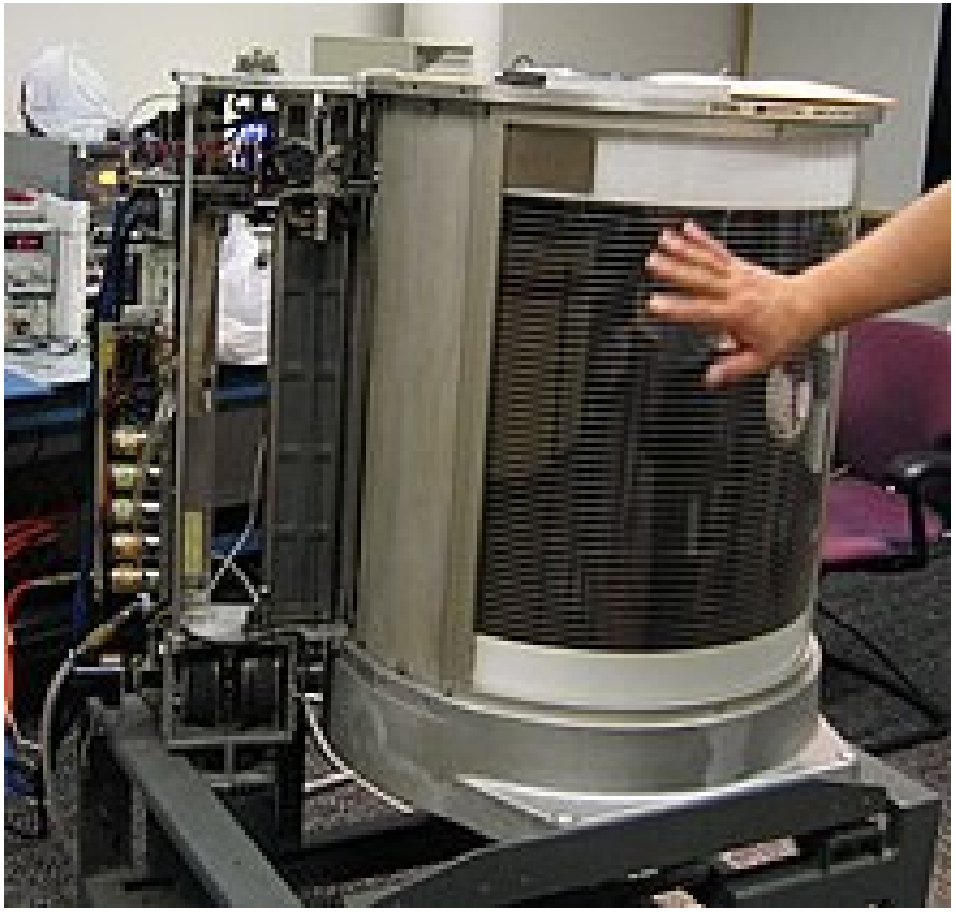
Historie

První pevný disk 1956

Parametry

- 4,2 MB
- 24 palců
- 78 ot./min
- 35 000 USD

IBM 305 RAMAC



Výhody x nevýhody

Výhody

- Poměr kapacita – cena
- Vysoká rychlost čtení/zápis
- Trvanlivost

Nevýhody

- Mechanické řešení
- Velký příkon energie
- Poškození
- Hmotnost

Princip uložení a parametry

Magnetický záznam

Magnetické plotny 1 – 5

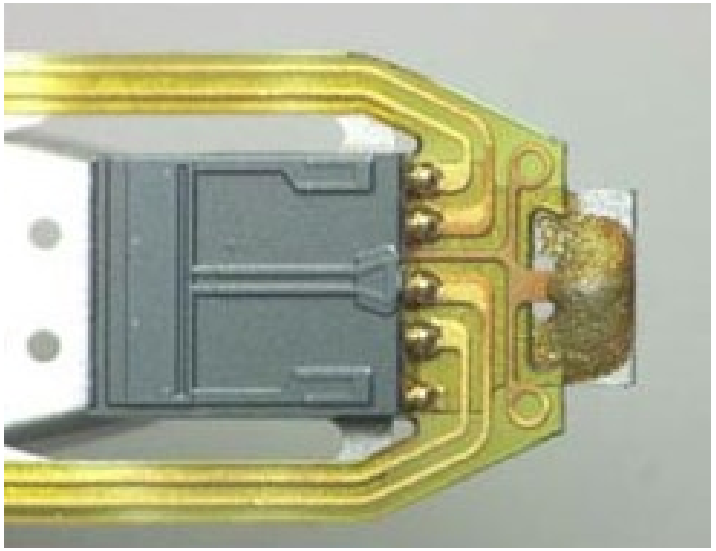
Otáčky 7200 – 15 000 ot./min

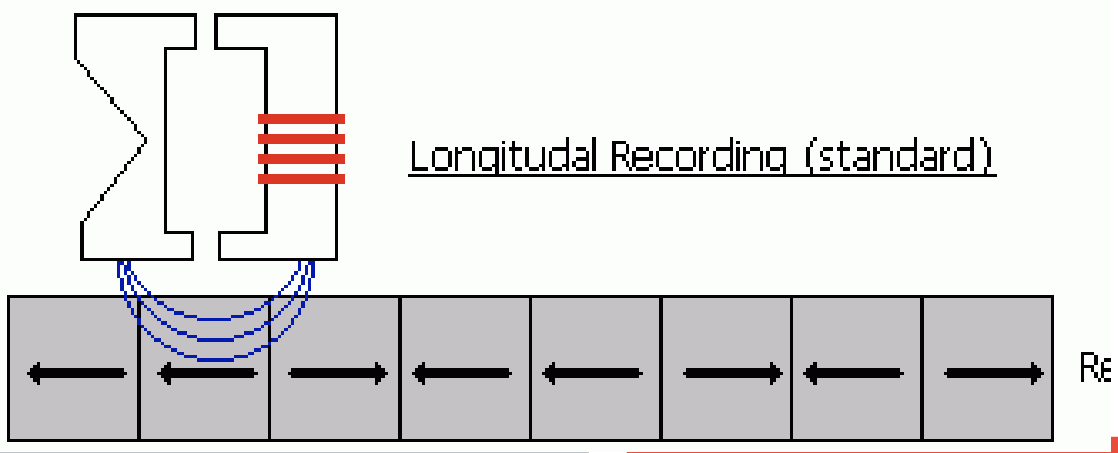
Kapacita

Hustota datového záznamu

2,5 – 3,5 palců

Princip uložení





ATA

(IDE, PATA)

Paralelní

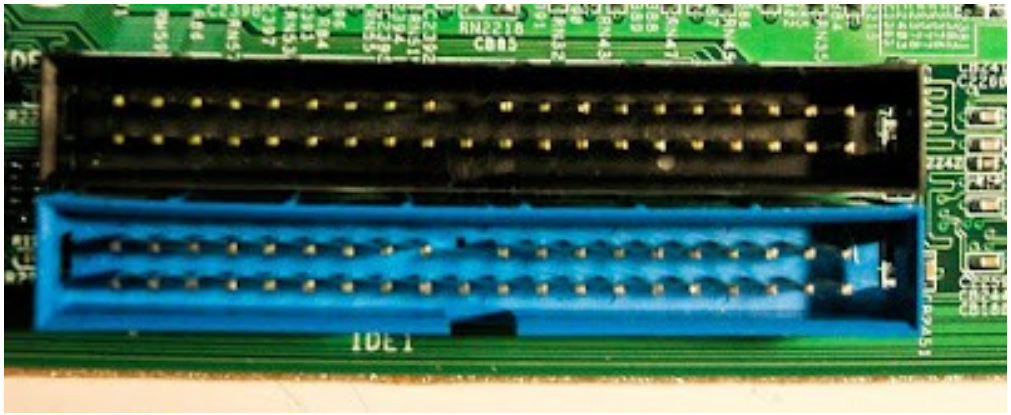
EIDE

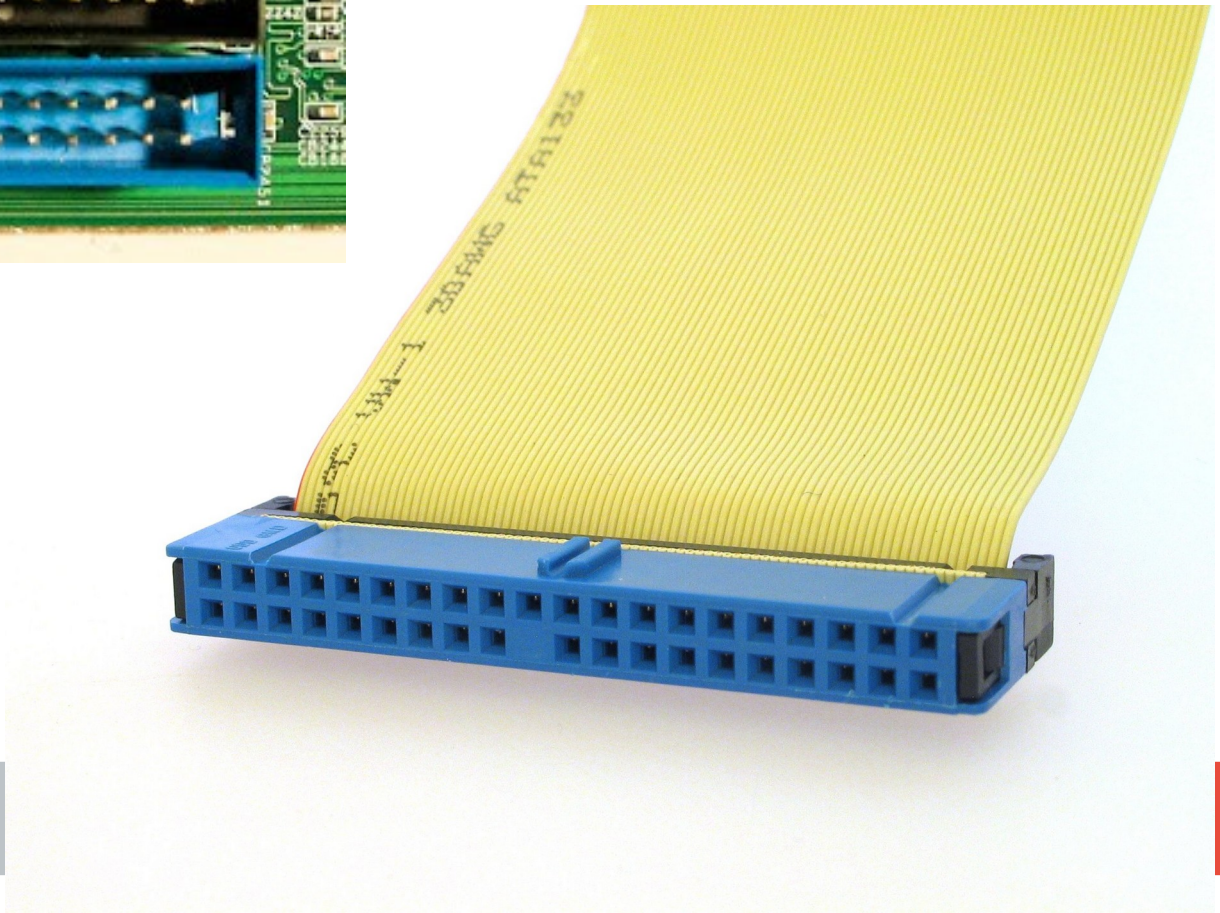
UltraATA

2 – 133 MB/s

DMA

ATA





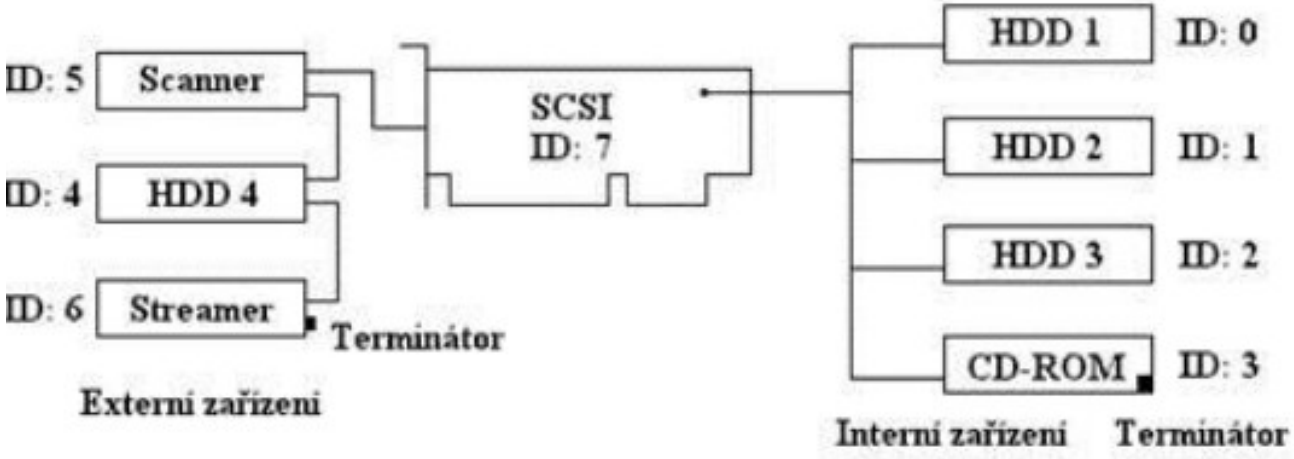
SCSI

Paralelní

Macintosh

Integrovaná

Přídavná karta



SATA

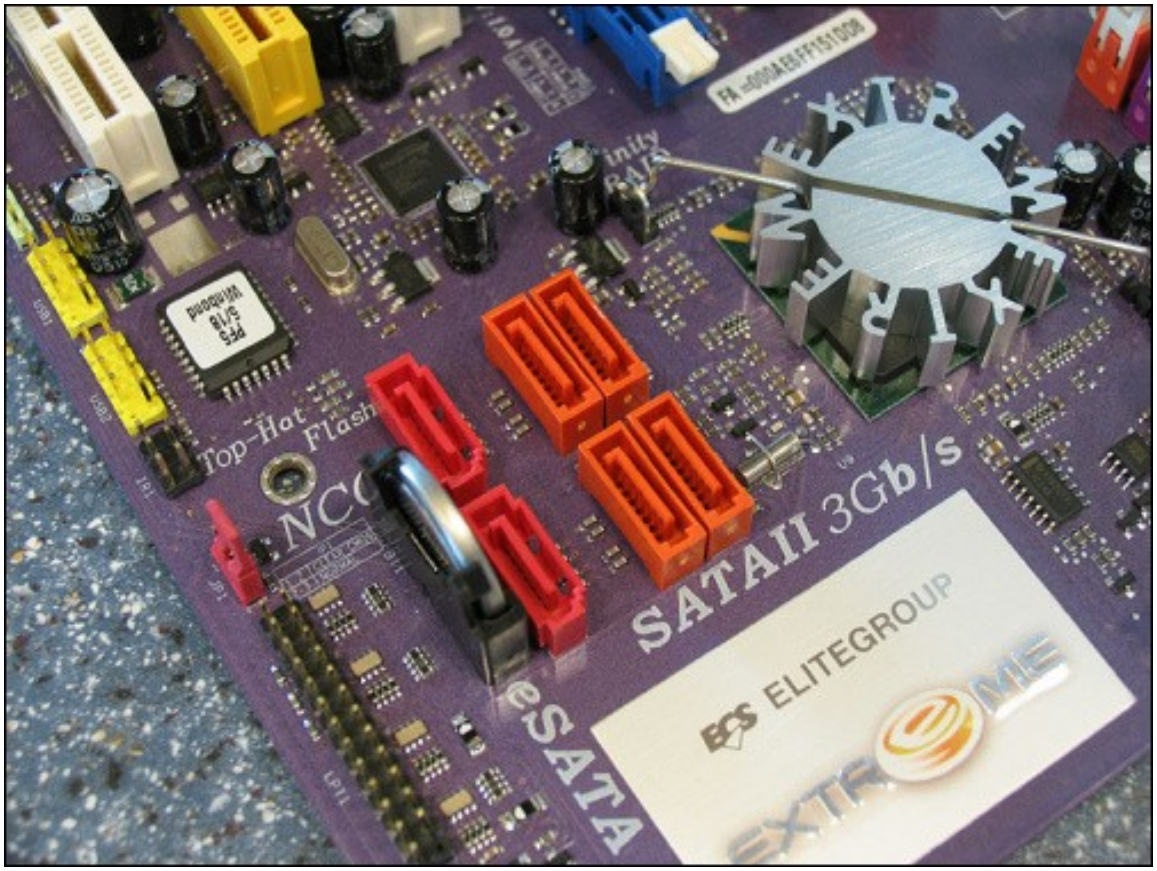
Sériové rozhraní

Kódování

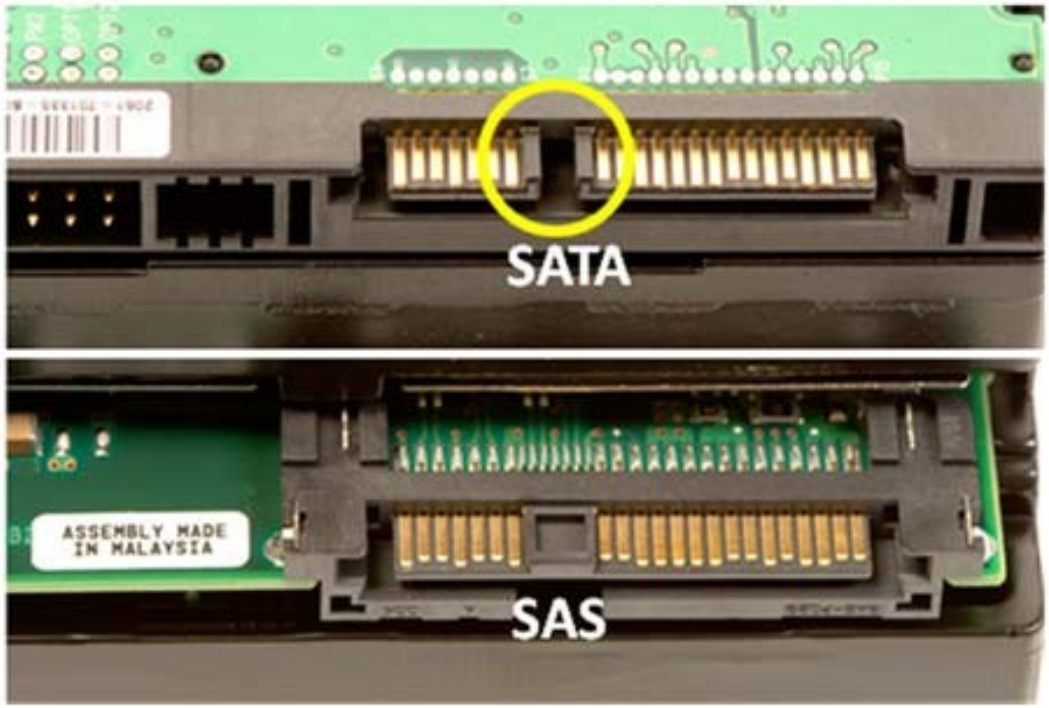
SATA

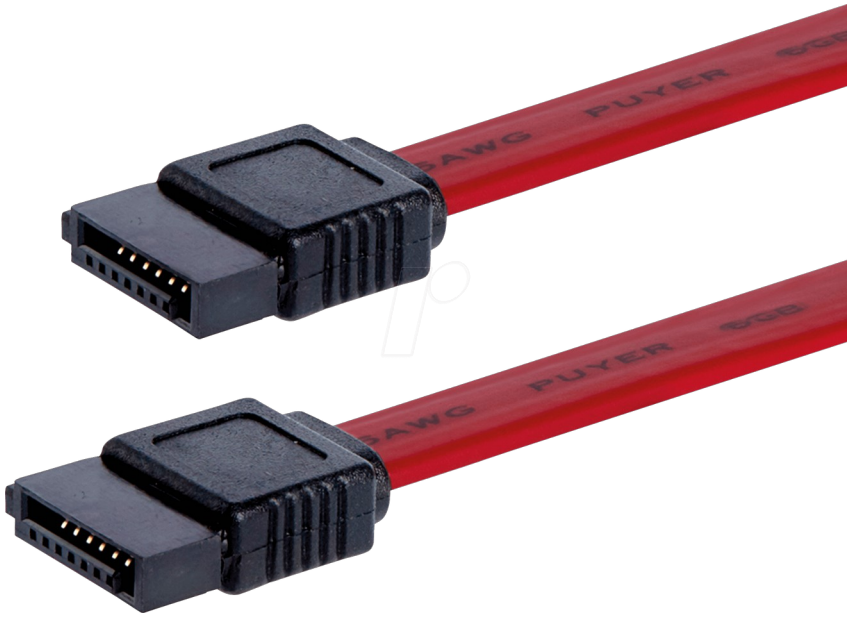
SATA II

SATA III



SATA



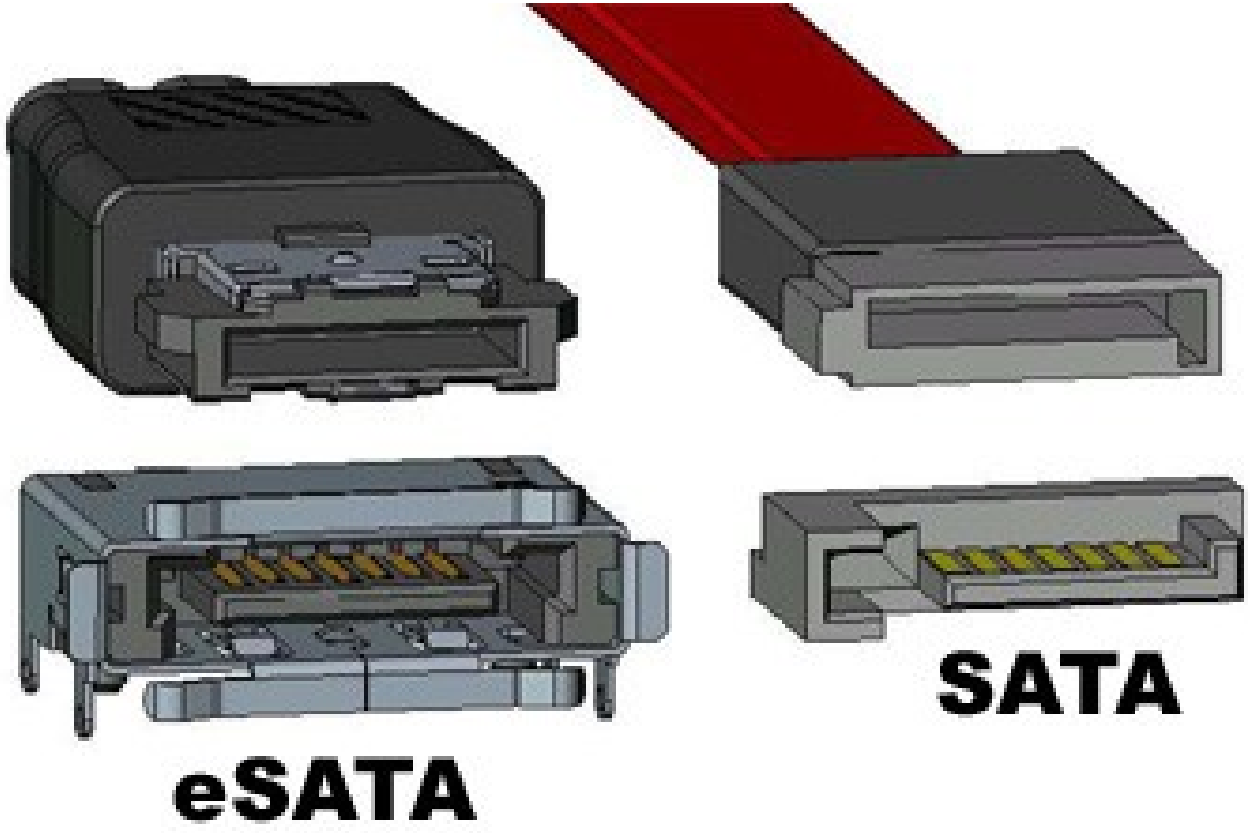


SATA

UMAX
2.5" SATA SSD
128GB
www.umax.cz

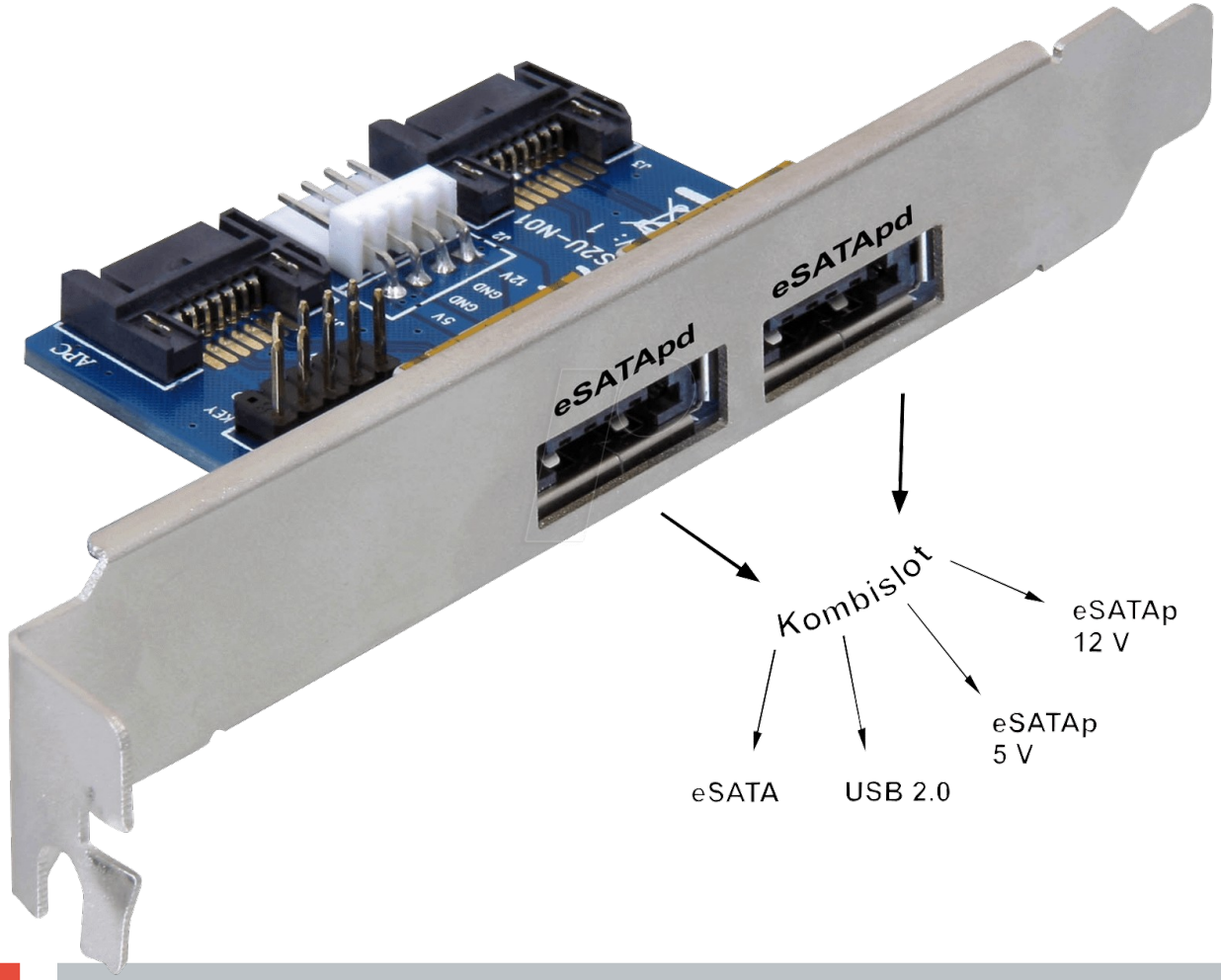
SECOMP
Einsteinstrasse 17-23
D-76275 Ettlingen
S/N: 191146 2
Made in China

eSATA, eSATAp

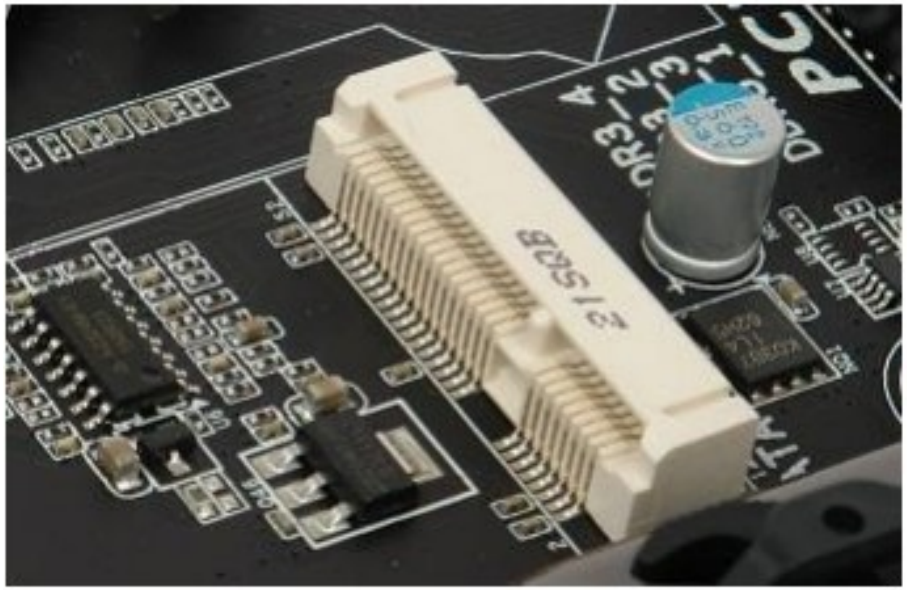


eSATA, eSATAp

eSATApd (Power Over eSATA Dual)



mSATA



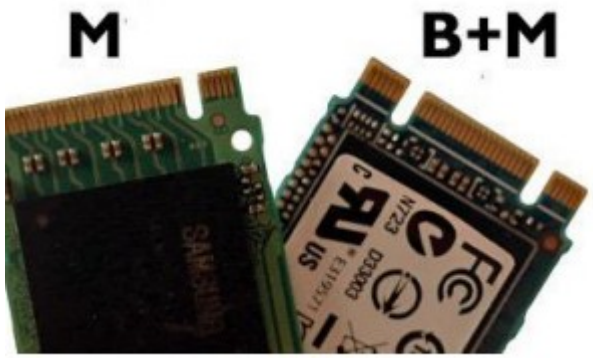
M.2

B – PCI Express x4 s CPU

4000 MB/s

B+M – SATA III

600 MB/s



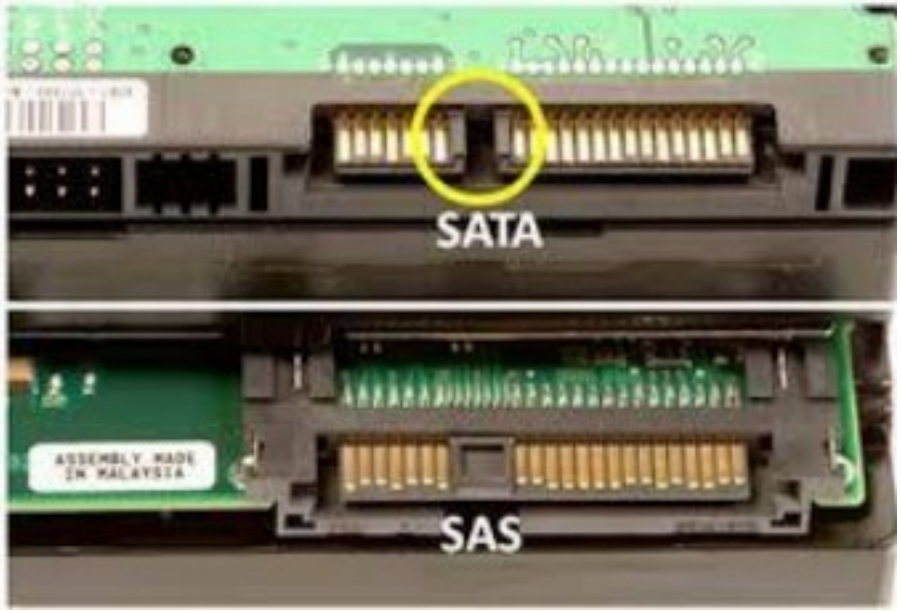
SAS

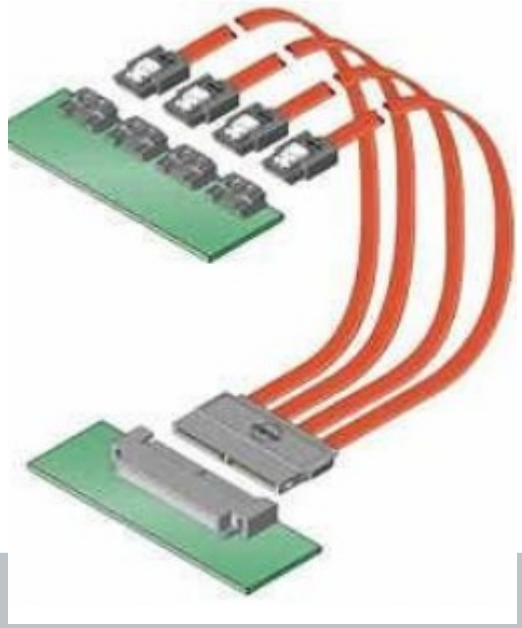
Sériové, nahrazuje SCSI

6 Gb/s

SAS – SATA

RAID





Optické mechaniky

CD-ROM

CD-RW

DVD

Blue-ray

Laser

Princip laserového světla

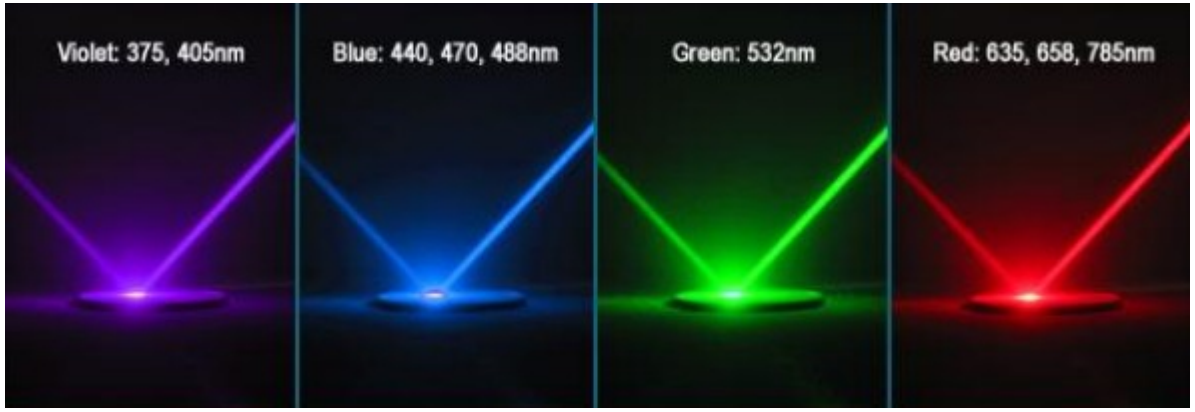
- Druhy médií
- Laserová hlava
- Optické čočky
- Fotodioda
- Princip ukládání – nepřepisovatelná media (vypalování dat)
- Princip ukládání – přepisovatelná media
- Čtení – digitální signál

Generace optických médií

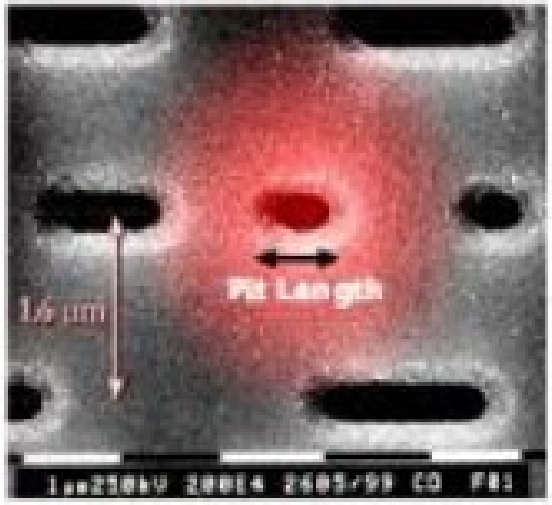
CD

DVD

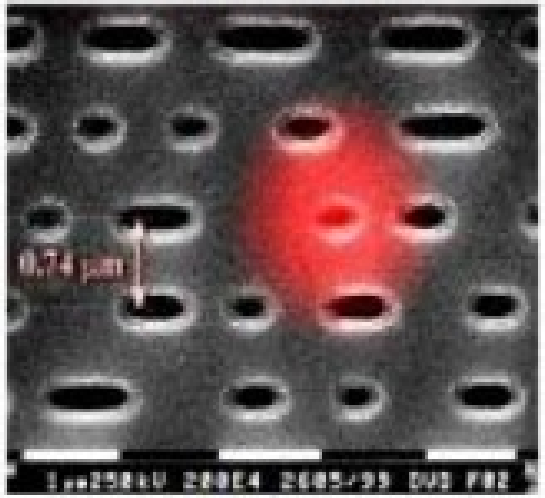
Blue-ray



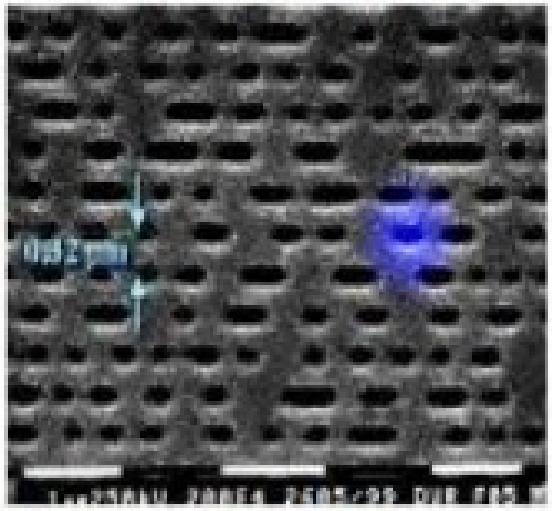
CD 0.7GB

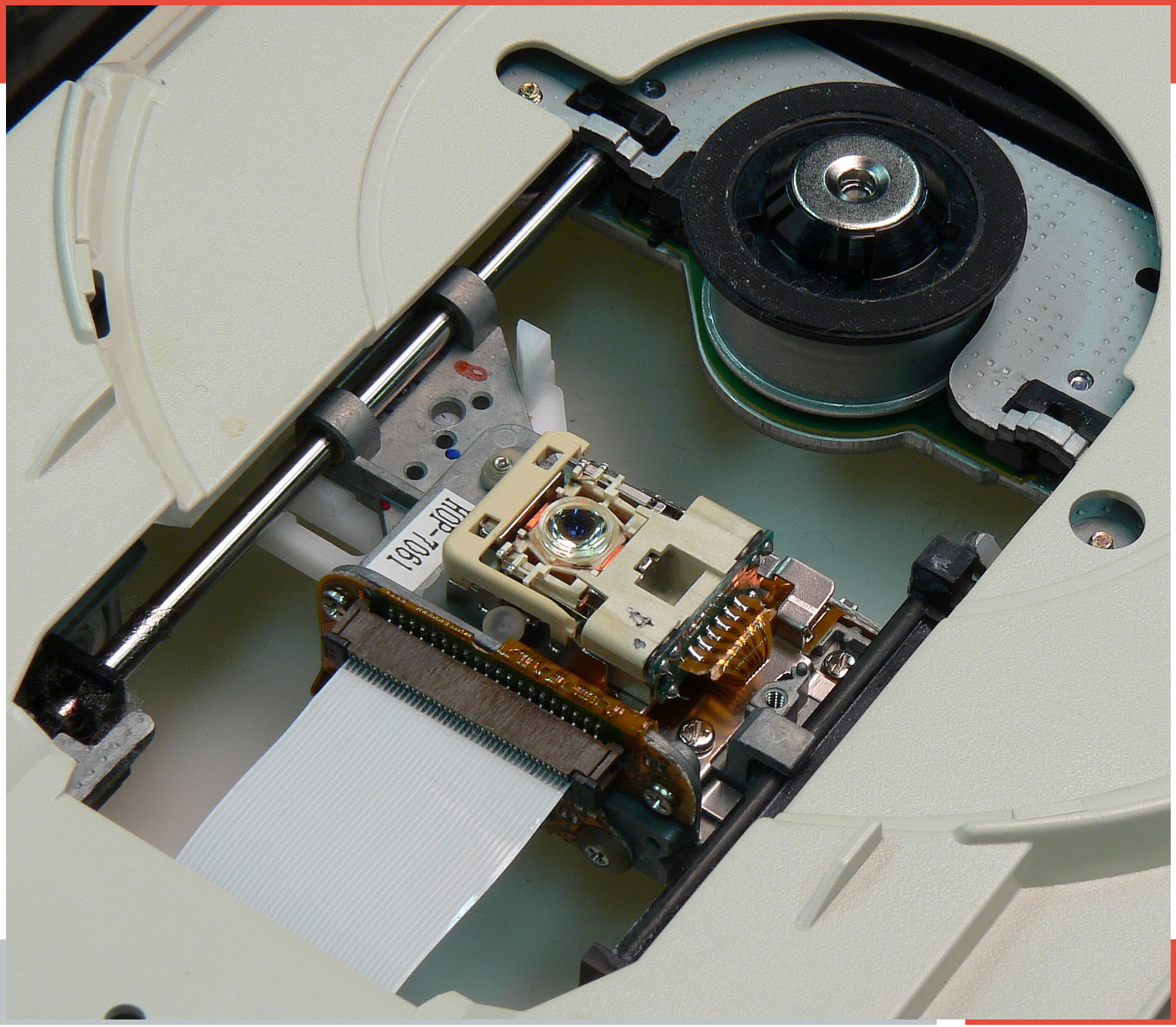


DVD 4.7GB



Blu-ray Disc 25GB




HOP-1061

Další parametry

Vrstvy

Rychlost 1x 150kB/s

Rozhraní